Master Comparison & Synthesis

Nine Theories of Consciousness in Dialogue

Convergences, Divergences, Empirical Evidence,
and the Road Ahead

Volume 10 of the Consciousness Theory Series

IIT • GWT • RPT • HOT • AST • PP • Orch OR • Panpsychism • CEMI

Comprehensive Study Material | February 2026

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Chapter 1

One-Page Theory Summaries

Theory	Proponents	Core Claim	Key Formalism	NCC Location
IIT	Tononi, Koch	Consciousness *is* integrated information (Φ)	Φ, cause-effect structure, EMD	Posterior cortex (hot zone)
GWT	Baars, Dehaene, Changeux	Consciousness = global broadcast of information	Ignition dynamics, Wilson-Cowan	Prefrontal-parietal network
RPT	Lamme	Consciousness = recurrent processing in sensory cortex	Laminar circuit models	Sensory cortex (V1+)
HOT	Rosenthal, Lau, Brown	Consciousness = higher-order representation of first-order state	Signal detection (d', meta-d')	Prefrontal cortex
AST	Graziano	Consciousness = brain's model of its own attention	Control theory, Kalman filter	TPJ, STS, mPFC
PP	Friston, Clark, Seth	Consciousness = best Bayesian guess (controlled hallucination)	Free energy, predictive coding	Hierarchical cortex
Orch OR	Penrose, Hameroff	Consciousness = quantum state reduction in microtubules	$\tau = \hbar/E_G$, OR	Microtubules (intraneuronal)
Pan.	Goff, Strawson, Chalmers	Consciousness is fundamental to all matter	$\Phi > 0$ (via IIT), Russellian monism	Everywhere / cosmos
CEMI	McFadden, Pockett	Consciousness = brain's EM field	Maxwell's eqs., Φ_{EM}	EM field (global)

Chapter 2

The Master Comparison Tables

2.1 Ontological Commitments

Theory	Metaphysics	Hard Problem	Qualia	Panpsychism?	Substrate Dep.?
IIT	Neutral monism	Addressed (identity)	Intrinsic to Φ	Yes ($\Phi > 0$)	Partially
GWT	Physicalism	Deferred	Functional	No	No
RPT	Physicalism	Deferred	Real (phenomenal)	No	Partially
HOT	Physicalism	Dissolved	Representational	No	No
AST	Physicalism	Dissolved (illusionist)	Illusory model	No	No
PP	Physicalism	"Real problem"	Model content	No (debated)	No
Orch OR	Proto-panpsychism	Addressed	In spacetime	Proto	Yes (quantum)
Pan.	Russellian monism	Dissolved	Fundamental	Yes (by def.)	Universal
CEMI	Physicalism	Relocated	Field patterns	No (debated)	Yes (EM field)

2.2 Neural Predictions

Theory	Key Brain Region	PFC Role	Key Signature	Attention Req.?	Phenomenal Overflow?
IIT	Posterior hot zone	Not required	Sustained posterior Φ	No	Yes
GWT	PFC-parietal	Essential	P3b, ignition, late (>300 ms)	Yes	No
RPT	Sensory cortex	Not required	Recurrent activity, VAN	No	Yes
HOT	PFC (dlPFC, vlPFC)	Essential	Late frontal activity	Not directly	No
AST	TPJ, STS, mPFC	Involved	Social cognition overlap	Indirectly	No
PP	Hierarchical cortex	Priors	Prediction error suppression	Via precision	Ambiguous
Orch OR	Microtubules	Irrelevant	Gamma (∼40 Hz)	No	N/A
Pan.	All matter	N/A	N/A	N/A	N/A
CEMI	EM field (global)	Via field	Coherent gamma field	Via synchrony	Possible

2.3 Implications for AI, Animals, and Clinical States

Theory	AI Consciousness?	Animal Consciousness	Disorders (VS/MCS)	Anesthesia
IIT	Only if high Φ (unlikely for feedforward nets)	Graded by Φ	VS = low Φ; MCS = intermediate	Disrupts integration
GWT	Yes (if has global workspace)	If has broadcast architecture	VS = no ignition; MCS = partial	Blocks broadcast
RPT	Yes (if has recurrence)	If has recurrent sensory processing	VS = no recurrence; MCS = partial	Blocks recurrence
HOT	Yes (if has higher-order reps)	Limited (needs meta-cognition)	VS = no HOR; MCS = degraded HOR	Blocks higher-order
AST	Yes (if models own attention)	If has attention schema	VS = no schema; MCS = degraded	Disrupts schema
PP	Yes (if has generative model)	Graded by model complexity	VS = no prediction; MCS = partial	Disrupts precision
Orch OR	No (classical AI cannot be conscious)	If has microtubule OR	VS = no OR; MCS = reduced OR	Disrupts tubulin
Pan.	Depends on variant	All organisms (graded)	All states have some Φ	Reduces Φ
CEMI	Only if has EM field feedback	If has coherent EM field	VS = incoherent field	Desynchronizes field

Chapter 3

Key Divergences

3.1 The Front vs. Back Debate

The single most empirically tractable disagreement among theories:

Front (PFC) vs. Back (Posterior Cortex)

Front theories (GWT, HOT, AST): Consciousness requires prefrontal cortex. The NCC includes frontal ignition, higher-order representations, or attention schemas in PFC.
Back theories (IIT, RPT): Consciousness resides in posterior sensory cortex (the "posterior hot zone"). PFC is involved in reporting and accessing conscious content but is not part of consciousness itself.
Empirical status: The COGITATE adversarial collaboration (2023) found evidence for *sustained posterior activity* (supporting IIT/RPT) but *weaker evidence for prefrontal ignition* (challenging GWT). However, frontal involvement was not entirely absent. Neither camp won decisively.
Resolution possibility: PFC may be required for *access* consciousness (reporting, acting on) but not for *phenomenal* consciousness (raw experience). If this distinction holds, both sides may be partially correct—but at different levels.

3.2 Phenomenal Overflow: Does it Exist?

Phenomenal Overflow

Overflow theorists (IIT, RPT): We experience more than we can report. Consciousness is richer than access—there is phenomenal experience of the entire visual field, even items we cannot report.
Anti-overflow theorists (GWT, HOT): If you can't report it, you weren't conscious of it. Phenomenal consciousness without access consciousness is incoherent or undetectable.
Key experiments: Sperling's partial report paradigm (1960) suggests subjects see the entire array but can only report a subset (supporting overflow). But critics argue this demonstrates iconic *memory*, not phenomenal consciousness.

3.3 Is Consciousness Computable?

Computable vs. Non-Computable

Computable (GWT, HOT, AST, PP, CEMI, RPT): Consciousness arises from information processing that can, in principle, be simulated on a Turing machine. AI consciousness is possible.
Non-computable (Orch OR): Consciousness requires non-algorithmic processes (objective reduction). No classical computer can be conscious, no matter how sophisticated.

Intermediate (IIT): Consciousness depends on *intrinsic causal structure*, not computation per se. A simulation of a brain on a classical computer would not be conscious (because the causal structure is in the computer hardware, not the simulation), but the right hardware *could* be conscious.

3.4 Identity vs. Correlation vs. Illusion

What is the Relationship Between Brain and Consciousness?

Identity: Consciousness *is* a specific physical process (IIT: Φ; CEMI: the EM field; Panpsychism: intrinsic nature of matter).
Functional correlation: Consciousness *arises from* certain computational operations (GWT: global broadcast; RPT: recurrent processing; PP: predictive inference).
Illusion/model: Consciousness as we conceive it doesn't exist as such; it is a model or representation (AST: attention schema; HOT: higher-order representation).

Chapter 4

Key Convergences

Despite deep disagreements, the nine theories converge on several important points:

Points of Agreement Across Theories

1. **Information integration is important.** IIT makes it central (Φ). GWT achieves it through broadcast. RPT through recurrence. CEMI through field superposition. PP through hierarchical inference. Even Orch OR proposes integration across microtubule networks. No theory denies that consciousness involves some form of information integration.
2. **Neural synchrony correlates with consciousness.** Gamma oscillations, thalamocortical coherence, and long-range synchrony are universally accepted correlates. GWT explains this via broadcast. RPT via recurrent loops. CEMI via field coherence. IIT via integrated cause-effect structure. The mechanism differs but the correlation is agreed upon.
3. **Anesthesia disrupts something fundamental.** All theories account for anesthetic abolition of consciousness—whether by disrupting integration (IIT), broadcast (GWT), recurrence (RPT), higher-order states (HOT), precision weighting (PP), field coherence (CEMI), or microtubule dynamics (Orch OR).
4. **Consciousness is graded, not binary.** All theories (except possibly simple versions of HOT) allow for degrees of consciousness—from minimal ($\Phi \approx 0$, simple organisms) to rich (human waking experience).
5. **Feedback/recurrence matters.** GWT requires recurrent broadcast. RPT is built on recurrence. PP requires top-down predictions (feedback). IIT requires bidirectional causal connections. CEMI requires field-neuron feedback. Even HOT implicitly requires feedback from higher-order to first-order areas.
6. **The posterior cortex is important for visual consciousness.** Even front-of-brain theories (GWT, HOT) acknowledge that visual experience involves posterior cortex. The debate is whether PFC is additionally required.

Chapter 5

The Adversarial Collaboration Results

5.1 The COGITATE Study (2023)

The largest adversarial collaboration in consciousness science tested predictions of IIT and GWT using pre-registered experiments:

COGITATE Results Summary

Design: Participants viewed visual stimuli while brain activity was recorded with fMRI, M/EEG, and ECoG across multiple sites. Theories pre-registered specific predictions.

IIT's predictions:

- Sustained activity in posterior cortex during conscious perception: **Partially supported.** Sustained posterior activity was observed, but the specific pattern did not perfectly match IIT's predictions.
- Content-specific information in posterior hot zone: **Supported.** Decoding of conscious content was strongest in posterior regions.
- PFC not necessary: **Partially supported.** PFC activity was less pronounced than GWT predicted, but was not entirely absent.

GWT's predictions:

- Late (>300 ms) "ignition" with prefrontal involvement: **Weakly supported.** Some late frontal activity observed but less robust than predicted.
- All-or-none (non-linear) transition: **Mixed.** Some evidence for non-linearity but not as sharp as predicted.
- P3b wave as marker of consciousness: **Partially supported.**

Overall verdict: Neither theory was clearly vindicated or refuted. The posterior cortex was more important than GWT predicted, and the PFC was less important than GWT predicted—tilting slightly toward IIT/RPT. But the results were not decisive enough to reject either theory.

5.2 Implications for All Nine Theories

Theory	COGITATE Impact	Implications
IIT	Partially supported	Posterior hot zone confirmed; sustained activity observed; but specific Φ-based predictions not directly tested
GWT	Partially challenged	Prefrontal ignition weaker than predicted; may need revision toward posterior emphasis
RPT	Indirectly supported	Posterior recurrence consistent with results; RPT not directly tested but compatible
HOT	Challenged	Frontal emphasis weakened; HOT may need to accommodate posterior consciousness
AST	Neutral	TPJ/STS not specifically tested; AST compatible with multiple outcomes
PP	Compatible	Hierarchical prediction error framework accommodates both frontal and posterior findings
Orch OR	Not tested	Microtubule-level phenomena not addressable by the experimental design
Pan.	Not tested	Metaphysical framework not empirically addressable
CEMI	Partially compatible	Coherent field patterns in posterior cortex consistent with predictions

5.3 Ongoing and Future Collaborations

The adversarial collaboration framework is expanding:

- **Phase 2:** Testing RPT and HOT alongside IIT and GWT with refined predictions.
- **Animal studies:** Extending the paradigms to non-human primates and potentially other species.
- **Clinical applications:** Testing theory predictions against disorders of consciousness (vegetative state, minimally conscious state).
- **No-report paradigms:** Experiments that detect consciousness without requiring subjects to report, addressing the access/phenomenal distinction directly.

Chapter 6

Compatibility and Complementarity

Many theories may be compatible because they operate at different levels of description:

6.1 A Multi-Level Framework

Theories as Layers

Level	Question	Theories
Metaphysical	What *is* consciousness?	Panpsychism, IIT (identity claim)
Mathematical	What are the formal principles?	IIT (Φ), PP (free energy), CEMI (Φ_{EM})
Algorithmic	What computation does consciousness perform?	PP (Bayesian inference), GWT (broadcast), HOT (meta-representation)
Architectural	What brain structure is needed?	GWT (workspace network), RPT (recurrent circuits)
Biophysical	What physical mechanism implements it?	CEMI (EM field), Orch OR (quantum biology)
Cognitive/functional	What role does consciousness play?	AST (attention control), HOT (metacognition)

A complete theory of consciousness may need elements from *multiple* levels. The theories are not necessarily competitors but may be complementary descriptions of the same phenomenon at different scales.

6.2 Natural Pairings

Compatible Theory Pairs

RPT + PP: Recurrent processing *is* the iterative prediction-error exchange in predictive coding. RPT describes the circuit architecture; PP describes the algorithm it implements.
GWT + CEMI: Global broadcast may be physically realized as a coherent EM field. The EM field provides the physical substrate; GWT describes the computational function.
IIT + Panpsychism: IIT's $\Phi > 0$ for simple systems provides a formal grounding for panpsychism. Panpsychism provides the metaphysical interpretation; IIT provides the math.

HOT + AST: The attention schema is a specific kind of higher-order representation. AST can be viewed as a biologically concrete implementation of HOT.
PP + GWT: Predictive processing describes the general algorithmic principle; global broadcast describes the specific architecture that makes some predictions conscious.
CEMI + IIT: The EM field may be the physical substrate that maximizes Φ through superposition-based integration.

Chapter 7

Scorecards

7.1 Theoretical Virtues

Theory	Math. Precision	Empirical Support	Falsifi-ability	Hard Problem	Clinical Utility	Parsi-mony
IIT	★★★	★★	★★	★★★	★★	★
GWT	★★	★★★	★★★	★	★★★	★★★
RPT	★★	★★★	★★★	★	★★	★★★
HOT	★★	★★	★★	★★	★	★★
AST	★	★★	★★	★★	★	★★★
PP	★★★	★★	★★	★★	★★	★★
Orch OR	★★	★	★★	★★★	★	★
Pan.	★	★	★	★★★	—	★★
CEMI	★★	★★	★★	★	★★	★★

★ = limited ★★ = moderate ★★★ = strong — = not applicable

7.2 Decisive Experiments

What experiment, if performed, would most strongly distinguish the theories?

Experiment	Would Support	Would Challenge
No-report paradigm: consciousness without frontal access	IIT, RPT	GWT, HOT
Full PFC lesion with preserved consciousness	IIT, RPT, AST	GWT, HOT
Specific Φ prediction confirmed	IIT	GWT (if posterior), HOT
AI with human-like consciousness	GWT, HOT, AST, PP	Orch OR (if classical)
Microtubule quantum coherence detected in vivo	Orch OR	All classical theories (mildly)
Disrupting EM field (not synapses) abolishes consciousness	CEMI	Circuit-only theories
Ephaptic-only communication supports consciousness	CEMI	Synaptic-only models

Chapter 8

Open Frontiers

8.1 Unsolved Problems

The Big Open Questions

1. **The Hard Problem:** No theory has truly solved why physical processes feel like anything. IIT and panpsychism claim to address it (by identity or fundamentality), but skeptics remain unconvinced. The gap between physics and phenomenology persists.
2. **The NCC vs. the mechanism:** We have many correlations but few causal explanations. Knowing *which* brain areas are involved doesn't explain *why* their activity is conscious.
3. **Phenomenal overflow:** Can there be consciousness without access? The answer would reshape the entire theoretical landscape. No-report paradigms are the best hope for resolution.
4. **The unity of consciousness:** How is the multiplicity of brain processes unified into a single experience? The binding problem remains unsolved at a deep level. IIT (exclusion), CEMI (field unity), and GWT (workspace unity) offer different answers.
5. **Animal and AI consciousness:** How do we determine which systems are conscious? Without solving the measurement problem (detecting consciousness from the outside), this question may remain unanswerable.
6. **The temporal structure of consciousness:** Is experience continuous or discrete? Does it come in "frames" (Orch OR: ∼25 ms; some evidence from temporal integration) or flow continuously?
7. **The combination problem:** If panpsychism is true, how do micro-experiences combine? If physicalism is true, how does experience emerge from non-experience? Both have a "combination problem."

8.2 Emerging Approaches

New Directions

1. **No-report and within-state paradigms:** Experimental designs that detect consciousness without requiring behavioral report, directly testing the access/phenomenal distinction.
2. **Perturbational complexity (PCI):** Theory-neutral measures of consciousness level based on brain responses to TMS. Already clinically useful and may discriminate between theories.
3. **Adversarial collaborations Phase 2+:** Expanding to test RPT, HOT, and potentially PP and AST with pre-registered predictions.
4. **Computational modeling:** Large-scale brain simulations that implement different theories and test whether they produce the right neural signatures.
5. **AI consciousness:** As AI systems grow more sophisticated, the question of machine consciousness becomes urgent. Theories make sharply different predictions here.
6. **Psychedelics research:** The REBUS model (PP) and altered states provide a rich testing

ground. Different theories make different predictions about why psychedelics alter consciousness.

7. **Cross-species neuroscience:** Testing theory predictions in diverse species (birds, cephalopods, insects) to determine the phylogenetic distribution of consciousness.
8. **Theory unification:** Attempts to combine compatible theories into multi-level frameworks that address multiple aspects of consciousness simultaneously.

8.3 Toward a Unified Theory?

A Speculative Synthesis

The nine theories may not be competing answers to the same question but rather answers to *different* questions about consciousness:

- **What is consciousness?** → Panpsychism / IIT (ontological question)
- **What are its mathematical principles?** → IIT (Φ) / PP (free energy)
- **What algorithm does it implement?** → PP (Bayesian inference) / GWT (broadcast)
- **What neural architecture does it need?** → GWT (workspace) / RPT (recurrence)
- **What physical mechanism implements it?** → CEMI (EM field) / Orch OR (quantum)
- **What is its cognitive function?** → AST (attention control) / HOT (metacognition)
- **Why does it exist?** → Still unanswered.

A mature science of consciousness will likely draw from multiple theories, integrating formal mathematical frameworks (IIT, PP) with specific neural mechanisms (GWT, RPT, CEMI) and cognitive functions (HOT, AST), grounded in a metaphysics that takes the Hard Problem seriously (Panpsychism, or a successor). The field is moving from an era of *theory proliferation* to one of *theory testing and integration*.

Chapter 9

Series Reference

The Complete Consciousness Theory Series

Vol. 0 Introduction to Consciousness Studies — Foundations, problems, methods, and the landscape of theories.

Vol. 1 Integrated Information Theory (IIT) — Axioms, postulates, Φ, qualia space, cause-effect structure, EMD, IIT 4.0.

Vol. 2 Global Workspace Theory (GWT) — Global broadcast, ignition dynamics, prefrontal-parietal workspace, Wilson-Cowan equations.

Vol. 3 Recurrent Processing Theory (RPT) — Three-stage framework, laminar circuitry, phenomenal overflow, VAN vs. DAN.

Vol. 4 Higher-Order Theories (HOT) — Rosenthal's HOT, perceptual reality monitoring, SDT, meta-d', metacognitive efficiency.

Vol. 5 Attention Schema Theory (AST) — Attention modeling, schema dynamics, TPJ/STS, evolutionary stages, social cognition.

Vol. 6 Predictive Processing & Active Inference (PP) — Free energy principle, variational inference, Markov blankets, hierarchical predictive coding.

Vol. 7 Orchestrated Objective Reduction (Orch OR) — Quantum mechanics, Diósi-Penrose criterion, microtubules, Gödel's theorems, spin networks.

Vol. 8 Panpsychism & Cosmopsychism — Russellian monism, combination problem, IIT as formal panpsychism, cosmopsychist dissolution.

Vol. 9 CEMI / EM Field Theories — Maxwell's equations, ephaptic coupling, field integration, CEMI download principle.

Vol. 10 Master Comparison & Synthesis (this document) — Cross-theory comparison tables, adversarial collaboration results, convergences, divergences, open frontiers.

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End of Series